

# Statistical Methods I: Linear Model

## Lec 03: Inference In Simple Linear Regression

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# Section 1

## Intro

## Review: Lecture 01

- In lecture 01, we introduced the simple linear regression model and the least squares estimation (LSE) method.
- LSE:

$$\text{SSE} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

$$\{\hat{\beta}_0, \hat{\beta}_1\} = \arg \min_{\beta_0, \beta_1} \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

- The LSEs are

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

## Review: Lecture 02

- In lecture 02, we discussed distribution, confidence interval, and hypothesis testing related to  $\beta_1$ . In particular, we showed that

- 

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x}) y_i}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

- $Var(\hat{\beta}_1) = \frac{\sigma^2}{S_{XX}}$ , where  $S_{XX} = \sum_{i=1}^n (x_i - \bar{x})^2$

- 

$$T = \frac{\hat{\beta}_1 - \beta_1}{s/\sqrt{S_{XX}}} = \frac{\hat{\beta}_1 - \beta_1}{se(\hat{\beta}_1)} \sim t_{n-2}$$

where  $se(\hat{\beta}_1) = \sqrt{\hat{Var}(\hat{\beta}_1)} = \sqrt{\frac{s^2}{S_{XX}}}$  and  $s^2 = \frac{SSE}{n-2}$ .

# Outline

- In this lecture, we will continue to discuss inference. We will focus on
  - estimating a mean response
  - predicting a new response
  - analysis of variance (ANOVA)

## Section 2

### Distribution of $\hat{\beta}_0$

# Distribution of $\hat{\beta}_0$

- Recall the simple linear regression model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \quad \epsilon_i \stackrel{iid}{\sim} N(0, \sigma^2), \quad i = 1, 2, \dots, n.$$

- The least squares estimator of  $\beta_0$  is

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

- The expectation of  $\hat{\beta}_0$  is

$$\begin{aligned} E(\hat{\beta}_0) &= E(\bar{y} - \hat{\beta}_1 \bar{x}) \\ &= E(\bar{y}) - \bar{x} E(\hat{\beta}_1) \\ &= \beta_0 + \beta_1 \bar{x} - \bar{x} \beta_1 = \beta_0. \end{aligned}$$

# Distribution of $\hat{\beta}_0$

- $\hat{\beta}_0$  is a linear function of  $y_i$ 's:

$$\begin{aligned}
 \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\
 &= \frac{1}{n} \sum_{i=1}^n y_i - \frac{\bar{x}}{S_{XX}} \sum_{i=1}^n (x_i - \bar{x}) y_i \\
 &= \sum_{i=1}^n \left( \frac{1}{n} - \frac{\bar{x}}{S_{XX}} (x_i - \bar{x}) \right) y_i.
 \end{aligned}$$

- Since  $\hat{\beta}_0$  is a linear combination of  $y_i$ 's, it is normally distributed.
- We have already find its mean (the LSE of  $\beta_0$  is unbiased). Next, we will find its variance.

## Variance of $\hat{\beta}_0$

- The variance of  $\hat{\beta}_0$  is

$$\begin{aligned}\text{Var}(\hat{\beta}_0) &= \text{Var} \left( \sum_{i=1}^n \left( \frac{1}{n} - \frac{\bar{x}}{S_{XX}} (x_i - \bar{x}) \right) y_i \right) \\ &= \sum_{i=1}^n \left( \frac{1}{n} - \frac{\bar{x}}{S_{XX}} (x_i - \bar{x}) \right)^2 \text{Var}(y_i) \\ &= \sigma^2 \sum_{i=1}^n \left( \frac{1}{n} - \frac{\bar{x}}{S_{XX}} (x_i - \bar{x}) \right)^2.\end{aligned}$$

- After some algebra, we have

$$\text{Var}(\hat{\beta}_0) = \sigma^2 \left( \frac{1}{n} + \frac{\bar{x}^2}{S_{XX}} \right).$$

## Distribution of $\hat{\beta}_0$

- We have shown that  $\hat{\beta}_0$  is a linear combination of  $y_i$ 's, so it is normally distributed. We have also found its mean and variance. Therefore,

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{XX}}\right)\right).$$

- When inference of  $\hat{\beta}_0$  is needed, we can construct a t-statistic.
- The t-statistic for  $\beta_0$  is

$$t = \frac{\hat{\beta}_0 - \beta_0}{s \sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{XX}}}} \sim t_{n-2},$$

where  $s^2 = \text{SSE}/(n-2)$  is the unbiased estimator of  $\sigma^2$ .

- Confidence interval and hypothesis testing for  $\beta_0$  can be conducted based on the above t-statistic.

## Variance of $\hat{\beta}_0$ : Method 2

- We present a different way to find  $\text{Var}(\hat{\beta}_0)$ .

$$\begin{aligned}\text{Var}(\hat{\beta}_0) &= \text{Var}(\bar{y} - \hat{\beta}_1 \bar{x}) \\ &= \text{Var}(\bar{y}) + \bar{x}^2 \text{Var}(\hat{\beta}_1) - 2\bar{x} \text{Cov}(\bar{y}, \hat{\beta}_1).\end{aligned}$$

- Need to find  $\text{Var}(\bar{y})$  and  $\text{Cov}(\bar{y}, \hat{\beta}_1)$ .
- Variance of  $\bar{y}$ :

$$\text{Var}(\bar{y}) = \text{Var}\left(\frac{1}{n} \sum_{i=1}^n y_i\right) = \frac{1}{n^2} n \sigma^2 = \sigma^2 / n$$

## Covariance of $\bar{y}$ and $\hat{\beta}_1$

$$\begin{aligned}
 \text{Cov}(\bar{y}, \hat{\beta}_1) &= \text{Cov}\left(\frac{1}{n} \sum_{i=1}^n y_i, \frac{1}{S_{XX}} \sum_{j=1}^n (x_j - \bar{x}) y_j\right) \\
 &= \frac{1}{n S_{XX}} \sum_{i=1}^n \sum_{j=1}^n (x_j - \bar{x}) \text{Cov}(y_i, y_j) \\
 &= \frac{1}{n S_{XX}} \sum_{i=1}^n \left[ (x_i - \bar{x}) \text{Cov}(y_i, y_i) + \sum_{j \neq i} (x_j - \bar{x}) \underbrace{\text{Cov}(y_i, y_j)}_{=0} \right] \quad (\text{since}) \\
 &= \frac{1}{n S_{XX}} \sum_{i=1}^n (x_i - \bar{x}) \text{Var}(y_i) \\
 &= \frac{\sigma^2}{n S_{XX}} \sum_{i=1}^n (x_i - \bar{x}) = 0.
 \end{aligned}$$

## Variance of $\hat{\beta}_0$ : Method 2

Putting everything together, we have

$$\begin{aligned}
 \text{Var}(\hat{\beta}_0) &= \text{Var}(\bar{y}) + \bar{x}^2 \text{Var}(\hat{\beta}_1) - 2\bar{x} \text{Cov}(\bar{y}, \hat{\beta}_1) \\
 &= \frac{\sigma^2}{n} + \bar{x}^2 \frac{\sigma^2}{S_{XX}} - 2\bar{x} \cdot 0 \\
 &= \sigma^2 \left( \frac{1}{n} + \frac{\bar{x}^2}{S_{XX}} \right).
 \end{aligned}$$

This is the same result as before.

## Section 3

$$\text{Cov}(\hat{\beta}_0, \hat{\beta}_1)$$

## Covariance of $\hat{\beta}_0$ and $\hat{\beta}_1$

- In many inference problems, we need the covariance of  $\hat{\beta}_0$  and  $\hat{\beta}_1$ . With normality assumptions, we can find the joint distribution of  $\hat{\beta}_0$  and  $\hat{\beta}_1$ .
- Covariance of  $\hat{\beta}_0$  and  $\hat{\beta}_1$ :

$$\begin{aligned}\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= \text{Cov}(\bar{y} - \hat{\beta}_1 \bar{x}, \hat{\beta}_1) \\ &= \text{Cov}(\bar{y}, \hat{\beta}_1) - \bar{x} \text{Var}(\hat{\beta}_1) \\ &= 0 - \bar{x} \frac{\sigma^2}{S_{XX}} \quad (\text{from previous results}) \\ &= -\frac{\bar{x} \sigma^2}{S_{XX}}.\end{aligned}$$

## Section 4

### Mean Response

# Mean Response

- A common interest in regression analysis is to estimate the mean response at a given value of  $x$ , say  $x_h$ .
- The mean response at  $x_h$  is

$$\mu_{y|x_h} = E(y|x = x_h) = \beta_0 + \beta_1 x_h.$$

- The point estimator of  $\mu_{y|x_h}$  is

$$\hat{\mu}_{y|x_h} = \hat{\beta}_0 + \hat{\beta}_1 x_h.$$

# Distribution of $\hat{\mu}_{y|x_h}$

- Since  $\hat{\mu}_{y|x_h}$  is a linear combination of  $\hat{\beta}_0$  and  $\hat{\beta}_1$ , therefore a linear combination of  $y_i$ 's (which are independently normal). Thus,  $\hat{\mu}_{y|x_h}$  is normally distributed.
- The expectation of  $\hat{\mu}_{y|x_h}$  is

$$\begin{aligned}
 E(\hat{\mu}_{y|x_h}) &= E(\hat{\beta}_0 + \hat{\beta}_1 x_h) \\
 &= E(\hat{\beta}_0) + x_h E(\hat{\beta}_1) \\
 &= \beta_0 + x_h \beta_1 = \mu_{y|x_h}.
 \end{aligned}$$

# Distribution of $\hat{\mu}_{y|x_h}$

- The variance of  $\hat{\mu}_{y|x_h}$  is

$$\begin{aligned}\text{Var}(\hat{\mu}_{y|x_h}) &= \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_h) \\ &= \text{Var}(\hat{\beta}_0) + x_h^2 \text{Var}(\hat{\beta}_1) + 2x_h \text{Cov}(\hat{\beta}_0, \hat{\beta}_1).\end{aligned}$$

- Plugging in the previous results, we have

$$\begin{aligned}\text{Var}(\hat{\mu}_{y|x_h}) &= \sigma^2 \left( \frac{1}{n} + \frac{\bar{x}^2}{S_{XX}} \right) + x_h^2 \frac{\sigma^2}{S_{XX}} + 2x_h \left( -\frac{\bar{x}\sigma^2}{S_{XX}} \right) \\ &= \sigma^2 \left( \frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_{XX}} \right).\end{aligned}$$

- Therefore,

$$\hat{\mu}_{y|x_h} \sim N \left( \mu_{y|x_h}, \sigma^2 \left( \frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_{XX}} \right) \right).$$

## Confidence Interval for Mean Response

- Since  $\hat{\mu}_{y|x_h}$  is normally distributed, we can construct a t-statistic for  $\mu_{y|x_h}$ :

$$t = \frac{\hat{\mu}_{y|x_h} - \mu_{y|x_h}}{s \sqrt{\frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_{XX}}}} \sim t_{n-2},$$

where  $s^2 = \text{SSE}/(n-2)$  is the unbiased estimator of  $\sigma^2$ .

- A  $(1 - \alpha)100\%$  confidence interval for  $\mu_{y|x_h}$  is

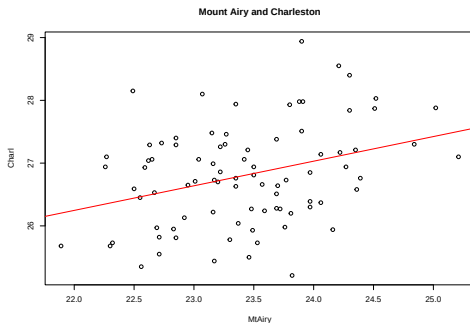
$$\hat{\mu}_{y|x_h} \pm t_{\alpha/2, n-2} \cdot s \sqrt{\frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_{XX}}},$$

where  $t_{\alpha/2, n-2}$  is the  $(1 - \alpha/2)$  quantile of the  $t$  distribution with  $n - 2$  degrees of freedom.

## Example: Mount Airy and Charleston

Summer temperature averages in Mount Airy, NC and Charleston, SC (1937-2021)

```
mtair=read.csv("data/mta1.csv", head=TRUE)
plot(mtair$MtAiry, mtair$Charl,
     xlab="MtAiry",
     ylab="Charl",
     main="Mount Airy and Charleston")
fit <- lm(Charl ~ MtAiry, data = mtair)
abline(fit, col="red", lwd=2)
```



- We are interested in a mean response at  $x_h = 23$  (Charleston's average summer temperature).

# Mean response at $x_h = 23$

```
x_h = 23
newdata <- data.frame(MtAiry = x_h)
#### (1) Using predict()
ci_est <- predict(fit, newdata, interval="confidence", level=0.95)
ci_est
```

```
##          fit          lwr          upr
## 1 26.63873 26.44618 26.83129
```

- Important: interval="confidence" should be used to produce the correct CI for a mean response.

# Mean response at $x_h = 23$

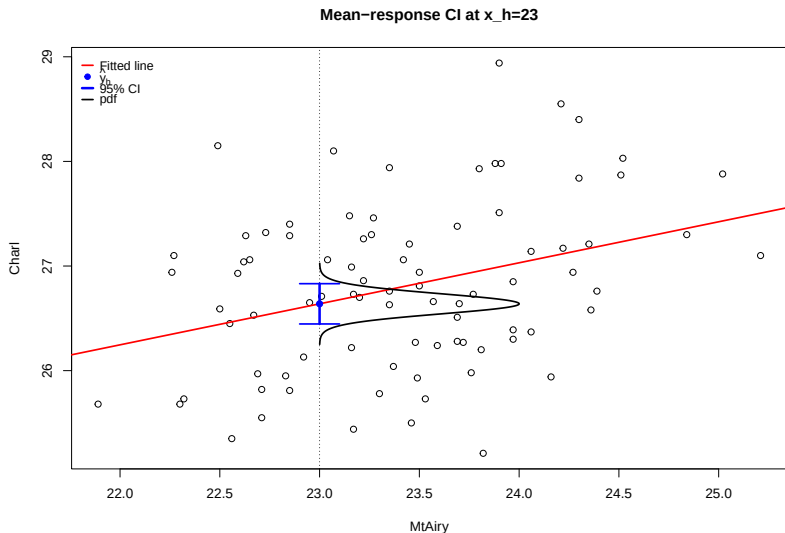
```
#### (2) Semi-manual: extract pieces from the fit and compute CI
n <- length(fit$fitted.values)      # should be 85
df <- n - length(coef(fit))         # = 83
tcrit <- qt(0.975, df)              # two-sided 95%

x      <- model.frame(fit)$MtAiry
yhat   <- coef(fit)[1] + coef(fit)[2]*x_h
s      <- summary(fit)$sigma        # residual std. error
xbar   <- mean(x)
Sxx    <- sum( (x - xbar)^2 )

se_mu_formula <- s * sqrt( 1/n + (x_h - xbar)^2 / Sxx )
ci_est_result <- data.frame(fit=yhat, lwr=yhat - tcrit*se_mu_formula, upr= yhat + tcrit*se_mu_formula)
row.names(ci_est_result) <- NULL
ci_est_result
```

```
##          fit          lwr          upr
## 1 26.63873 26.44618 26.83129
```

# Mean response at $x_h = 23$



## Section 5

### Prediction of a New Response

## Prediction of a New Response

- Another common interest in regression analysis is to predict a new response at a given value of  $x$ , say  $x_h$ .
- The new response at  $x_h$  is

$$y_{\text{new}} = \beta_0 + \beta_1 x_h + \epsilon,$$

where  $\epsilon \sim N(0, \sigma^2)$  is independent of the  $\epsilon_i$ 's in the original model.

- The point predictor of  $y_{\text{new}}$  is

$$\hat{y}_{\text{new}} = \hat{\beta}_0 + \hat{\beta}_1 x_h.$$

## Mean and Variance of $\hat{y}_{\text{new}}$

- Note that  $\hat{y}_{\text{new}}$  is the same as  $\hat{\mu}_{y|x_h}$ , the point estimator of the mean response at  $x_h$ .
- They have the same mean and variance.
- To construct a prediction interval for  $y_{\text{new}}$ , we need to account for the extra variability due to  $\epsilon$ .
- Technically, we need to find the distribution of

$$\hat{y}_{\text{new}} - y_{\text{new}}$$

# Distribution of $\hat{y}_{\text{new}} - y_{\text{new}}$

$$\begin{aligned}\hat{y}_{\text{new}} - y_{\text{new}} &= (\hat{\beta}_0 + \hat{\beta}_1 x_h) - (\beta_0 + \beta_1 x_h + \epsilon) \\ &= (\hat{\beta}_0 - \beta_0) + (\hat{\beta}_1 - \beta_1) x_h - \epsilon.\end{aligned}$$

- Mean:

$$\begin{aligned}E[\hat{y}_{\text{new}} - y_{\text{new}}] &= E(\hat{\beta}_0 - \beta_0) + x_h E(\hat{\beta}_1 - \beta_1) - E(\epsilon) \\ &= 0 + x_h \cdot 0 - 0 = 0.\end{aligned}$$

## Distribution of $\hat{y}_{\text{new}} - y_{\text{new}}$

- Variance:

$$\begin{aligned} \text{Var}(\hat{y}_{\text{new}} - y_{\text{new}}) &= \text{Var}(\hat{y}_{\text{new}}) + \text{Var}(y_{\text{new}}) + 0 \\ &= \sigma^2 \left( \frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_{XX}} \right) + \sigma^2 \\ &= \sigma^2 \left( \mathbf{1} + \frac{1}{n} + \frac{(x_h - \bar{x})^2}{S_{XX}} \right). \end{aligned}$$

The first step is true because  $y_{\text{new}}$  is a new data point, thus is independent of  $\hat{\mu}_{y|x_h}$ , which relies on the old data.

## Distribution of $\hat{y}_{\text{new}} - y_{\text{new}}$

- Note that the variance of  $\hat{y}_{\text{new}} - y_{\text{new}}$  has an extra  $\sigma^2$  term compared to the variance of  $\hat{\mu}_{y|x_h}$ .
- This extra  $\sigma^2$  term accounts for the variability of the new observation  $y_{\text{new}}$  due to the random error  $\epsilon$ .
- We already explained that  $\hat{\mu}_{y|x_h}$  follows a normal distribution.
- $y_{\text{new}}$  also follows a normal distribution and it is independent of  $\hat{\mu}_{y|x_h}$  because it is a new data point.

## Distribution of $\hat{y}_{\text{new}} - y_{\text{new}}$

- As a result,  $\hat{y}_{\text{new}} - y_{\text{new}}$  also follows a normal distribution.
- The standardized version is

$$\frac{\hat{y}_{\text{new}} - y_{\text{new}}}{\sigma \sqrt{1 + \frac{1}{n} + \frac{(x_h - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}}$$

which follows a standard normal distribution.

- But we don't know  $\sigma$ , so we need to estimate it by  $s = \sqrt{\text{SSE}/(n-2)}$ .
- The resulting distribution is a t-distribution with  $n-2$  degrees of freedom.

# Prediction Interval

- We explained that

$$t = \frac{\hat{Y}_{\text{new}} - Y_{\text{new}}}{s \sqrt{1 + \frac{1}{n} + \frac{(x_h - \bar{X})^2}{S_{XX}}}} \sim t_{n-2},$$

where  $s^2 = \text{SSE}/(n-2)$  is the unbiased estimator of  $\sigma^2$ .

- A  $(1 - \alpha)100\%$  prediction interval for  $Y_{\text{new}}$  is

$$\hat{Y}_{\text{new}} \pm t_{\alpha/2, n-2} \cdot s \sqrt{1 + \frac{1}{n} + \frac{(x_h - \bar{X})^2}{S_{XX}}},$$

where  $t_{\alpha/2, n-2}$  is the  $(1 - \alpha/2)$  quantile of the  $t$  distribution with  $n - 2$  degrees of freedom.

# Prediction at $x_h = 23$

```
x_h = 23
newdata <- data.frame(MtAiry = x_h)
#### (1) Using predict()
ci_pred <- predict(fit, newdata, interval="prediction", level=0.95)
ci_pred
```

```
##          fit          lwr          upr
## 1 26.63873 25.11861 28.15886
```

# Prediction at $x_h = 23$

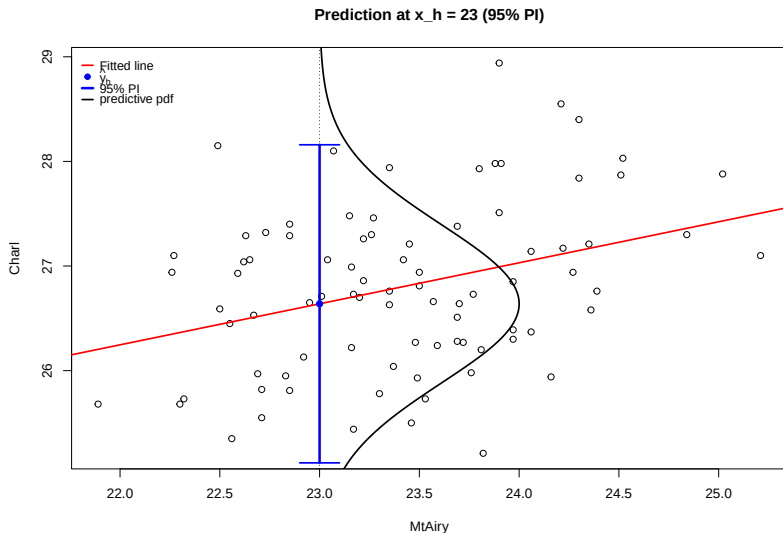
```
#### (2) Semi-manual: extract pieces from the fit and compute CI
n <- length(fit$fitted.values)      # should be 85
df <- n - length(coef(fit))         # = 83
tcrit <- qt(0.975, df)              # two-sided 95%

x      <- model.frame(fit)$MtAiry
yhat   <- coef(fit)[1] + coef(fit)[2]*x_h
s      <- summary(fit)$sigma        # residual std. error
xbar   <- mean(x)
Sxx    <- sum( (x - xbar)^2 )

se_mu_formula <- s * sqrt(1 + 1/n + (x_h - xbar)^2 / Sxx )
ci_pred_result <- data.frame(fit=yhat, lwr=yhat - tcrit*se_mu_formula, upr=yhat + tcrit*se_mu_formula)
row.names(ci_pred_result) <- NULL
ci_pred_result
```

```
##          fit          lwr          upr
## 1 26.63873 25.11861 28.15886
```

# Prediction at $x_h = 23$

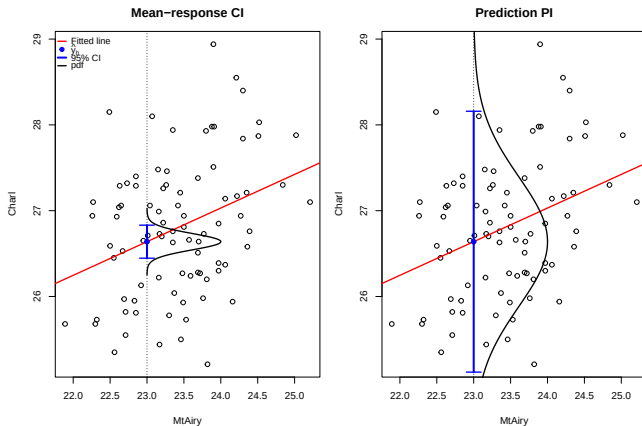


## Section 6

### Estimate vs Predict

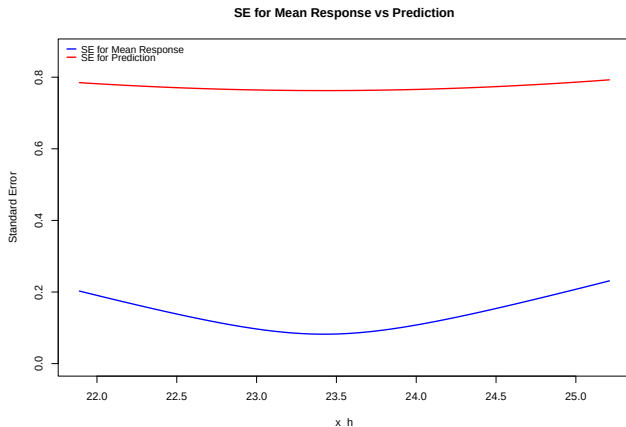
# Estimate a mean response vs. predict a new response

Mount Airy and Charleston: Mean vs Prediction at  $x_h = 23$

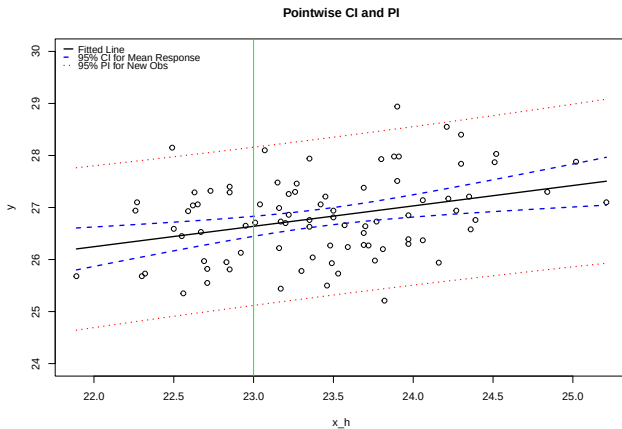


# Pointwise Curves

- The standard errors for mean response and prediction can be computed for a range of  $x_h$  values.



# Pointwise Curves for intervals



## Section 7

### ANOVA in SLR

# Sum of Squares Decomposition

- Total Sum of Squares (SST):

$$SST = \sum_{i=1}^n (y_i - \bar{y})^2$$

The total variation in the response variable.

- Regression Sum of Squares (SSR):

$$SSR = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

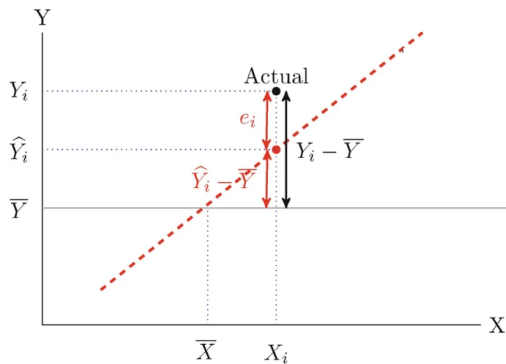
The variation explained by the predictor (regression model).

- Error Sum of Squares (SSE):

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

The variation not explained by the model (regression).

# Sums of Squares



# Decomposition of the Total Sum of Squares

It can be shown that:

$$SST = SSR + SSE$$

Proof:

$$\begin{aligned} SST &= \sum_{i=1}^n (y_i - \bar{y})^2 \\ &= \sum_{i=1}^n [(y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})]^2 \\ &= \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 + 2 \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) \\ &= SSE + SSR + 2 \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) \end{aligned}$$

The last term is zero because

$$\begin{aligned}
 \sum_{i=1}^n (y_i - \hat{y}_i)(\hat{y}_i - \bar{y}) &= \sum_{i=1}^n e_i(\hat{y}_i - \bar{y}) \\
 &= \sum_{i=1}^n e_i \hat{y}_i - \bar{y} \sum_{i=1}^n e_i \\
 &= \sum_{i=1}^n e_i \hat{y}_i - \bar{y} \cdot 0 \quad (\text{since } \sum e_i = 0) \\
 &= \sum_{i=1}^n e_i(\hat{\beta}_0 + \hat{\beta}_1 X_i) \\
 &= \hat{\beta}_0 \sum_{i=1}^n e_i + \hat{\beta}_1 \sum_{i=1}^n e_i X_i \\
 &= \hat{\beta}_0 \cdot 0 + \hat{\beta}_1 \cdot 0 \quad (\text{from normal equations}) \\
 &= 0.
 \end{aligned}$$

# ANOVA Table for SLR

An analysis of variance (ANOVA) table summarizes the results of the regression analysis. - SSTO is decomposed to SSR and SSE. - An F-test is used for assessing whether there is a linear trend between  $x$  and  $y$ :

$$H_0 : \beta_1 = 0 \text{ vs. } H_a : \beta_1 \neq 0.$$

- $R^2$  can be computed from the table. (how?)

| Source            | DF      | SS         | MS                  | F             |
|-------------------|---------|------------|---------------------|---------------|
| <i>Regression</i> | 1       | <i>SSR</i> | $MSR = SSR/1$       | $F = MSR/MSE$ |
| <i>Error</i>      | $n - 2$ | <i>SSE</i> | $MSE = SSE/(n - 2)$ |               |
| <i>Total</i>      | $n - 1$ | <i>SST</i> |                     |               |

# ANOVA Table for SLR

- DF: degrees of freedom
- SS: sum of squares
- MS: mean square (SS divided by corresponding DF)
- F: F-statistic for testing  $H_0 : \beta_1 = 0$  vs.  $H_a : \beta_1 \neq 0$
- Under  $H_0$ ,  $F \sim F_{1,n-2}$ .

## Expected Mean Squares

- We will rewrite SSR in terms of  $\hat{\beta}_1$ :

$$\begin{aligned} SSR &= \sum_{i=1}^n (\hat{y}_i - \bar{y})^2 \\ &= \sum_{i=1}^n (\hat{\beta}_0 + \hat{\beta}_1 X_i - \bar{y})^2 \\ &= \sum_{i=1}^n (\hat{\beta}_1 (X_i - \bar{X}))^2 \quad (\text{since } \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{X}) \\ &= \hat{\beta}_1^2 \sum_{i=1}^n (X_i - \bar{X})^2 = \hat{\beta}_1^2 S_{XX}. \end{aligned}$$

- Therefore,

$$MSR = SSR/1 = \hat{\beta}_1^2 S_{XX}.$$

## Expected Mean Squares

- We find that  $SSR = \hat{\beta}_1^2 S_{XX}$ . But  $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{XX})$ . Note that  $Var(\hat{\beta}_1) = E(\hat{\beta}_1^2) - E^2[\hat{\beta}_1]$ . Therefore,

$$\begin{aligned} E[SSR] &= E[\hat{\beta}_1^2 S_{XX}] \\ &= S_{XX} E[\hat{\beta}_1^2] \\ &= S_{XX} \left( Var(\hat{\beta}_1) + (E[\hat{\beta}_1])^2 \right) \\ &= S_{XX} \left( \frac{\sigma^2}{S_{XX}} + \beta_1^2 \right) \\ &= \sigma^2 + \beta_1^2 S_{XX}. \end{aligned}$$

$$\text{And } E[MSR] = E[SSR/1] = \sigma^2 + \beta_1^2 S_{XX}.$$

## Expected Mean Squares

- It can be shown that

$$E(MSE) = \sigma^2.$$

- When  $\beta_1 = 0$ ,  $E(MSR) = E(MSE) = \sigma^2$ .
- When  $\beta_1 \neq 0$ ,  $E(MSR) > E(MSE)$ .
- This justifies the F-test for testing  $H_0 : \beta_1 = 0$  vs.  $H_a : \beta_1 \neq 0$ .

# F Test

- For testing  $H_0 : \beta_1 = 0$  vs.  $H_a : \beta_1 \neq 0$ , the F-statistic is

$$F = \frac{MSR}{MSE}.$$

- We reject  $H_0$  if  $F$  is too large
  - The rejection region is  $F > F_{\alpha,1,n-2}$ , where  $F_{\alpha,1,n-2}$  is the  $(1 - \alpha)$  quantile of the  $F$  distribution with 1 and  $n - 2$  degrees of freedom.
- The p-value is  $P(F_{1,n-2} \geq F_{\text{obs}})$ , where  $F_{\text{obs}}$  is the observed value of the F-statistic.

## F test and t-test

- The F-statistic can be rewritten as

$$F = \frac{MSR}{MSE} = \frac{\hat{\beta}_1^2 S_{XX}}{s^2} = \left( \frac{\hat{\beta}_1}{s/\sqrt{S_{XX}}} \right)^2 = t^2,$$

where  $t = \hat{\beta}_1/(s/\sqrt{S_{XX}})$  is the t-statistic for testing  $H_0 : \beta_1 = 0$ .

- Thus, the F-test and the t-test for testing  $H_0 : \beta_1 = 0$  are equivalent in SLR.
- In fact, as  $F_{1,n-2}$  distribution is also equivalent to the distribution of the square of a random variable whose distribution is  $t_{n-2}$ , the F test is equivalent to the two-sided t test.

## Example: ANOVA Table for SLR

```
summary(fit)
```

```
##
## Call:
## lm(formula = Charl ~ MtAiry, data = mtair)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.7502 -0.6291 -0.0123  0.5585  1.9484
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   17.6208     2.8325   6.221 1.91e-08 ***
## MtAiry         0.3921     0.1209   3.244  0.0017 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7581 on 83 degrees of freedom
## Multiple R-squared:  0.1125, Adjusted R-squared:  0.1018
## F-statistic: 10.52 on 1 and 83 DF, p-value: 0.001701
```

## Example: ANOVA Table for SLR

```
anova(fit)
```

```
## Analysis of Variance Table
##
## Response: Charl
##           Df Sum Sq Mean Sq F value    Pr(>F)
## MtAiry      1  6.047   6.0470   10.521 0.001701 **
## Residuals  83 47.705   0.5748
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```